

$$M \quad P - Mg \frac{1}{n} - MF$$

$$P = M \left(\frac{g}{n} + F \right)$$

$$P = \frac{M}{F} \left(\frac{g}{n} + F \right)$$

$$P = \frac{M}{F} (nF + g)$$

$$P = \frac{M}{N} (g + nF)$$

Work done by the engine / second

$$= PV$$

$$= \frac{MV}{N} (g + nF) \quad \text{Foot-Poundals}$$

Let,

H be the required HP

then, work-Done

$$= H \times 550 \times g \quad \text{Ft-p/s}$$

therefore,

$$H \times 550 \times g = \frac{MV}{N} (g + nF)$$

$$H = \frac{\frac{MV}{N} (g + nF)}{550g} = \frac{MV (nF + g)}{550gh}$$

Q) A train of mass $1b$ which ascending a smooth incline of ~~one in N~~

of 1 in N at when the velocity of the train v ~~foot~~ ft/s , its acceleration f ft/s^2 Prove that,

the effective HP of Engine

$$\frac{Mv(nf+g)}{550gn} \quad \frac{Mv(nf+g)}{550gn}$$

→ Let, $\theta = 1/N$

α, β implies an angle

$\sin \theta$

$P =$ force ^{exerted} by the ^{Engine} ~~force~~ and Resistance is 0 .

Resultant acting on the mass of the train

$$= P - \cancel{mg \cos \theta}$$

$$\cancel{P - mg \sin \theta} \quad P - Mg \sin \theta$$

therefor [Ⓟ]

$$= P - Mg \sin \theta$$

$$= \cancel{P - Mg \sin \theta} \quad Mg \sin \theta - MF$$

Unit of work:

CGS System:

FPS -

MKS/SI -

~~CGS sy~~

Eng

Foot-poundal

Kilogram

$$1 \text{ Joule} = 10^7 \text{ ergs}$$

Hook's Law:

$$T = \lambda x/a$$

Power:

$$\text{Power} = F \times v$$

$$(P) = FV$$

$$1 \text{ Horse power} = 550 \text{ foot pound / second}$$

$$1 \text{ HP} = 746 \text{ Wat}$$

Energy:

Energy two types:

1) Kinetic Energy: (K.E)

$$K.E = \frac{1}{2} mv^2$$

2) Potential Energy:

$$P.E = mgh$$

Work done by a variable force

case: 1

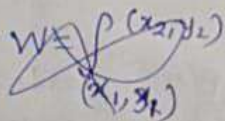
Work done by a variable force when the Displacement is along the line of action of the force.

$$\theta = 0$$

$$W = \int_a^b F(x) dx$$

case: 2

Work done by a variable force when the Displacement of the point of application along a plane curve.



$$W = \int_{(x_1, y_1)}^{(x_2, y_2)} x dx + y dy$$

$$= \int_A^B F \cdot ds$$

Energy

$$\text{Kinetic Energy} = \frac{1}{2}mv^2$$

$$\text{Potential} = mgh$$

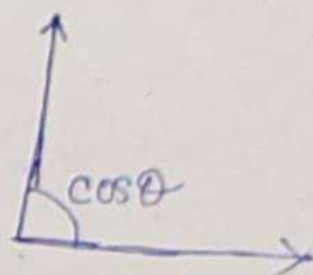
Date: 31.10.2025

Work: Work = Force $\times$ Displacement of the point of application along the line of action of force.

$$\text{WORK} = F \times S \cos \theta$$

$$\begin{aligned}\text{Work Done} &= \vec{F} \cdot \vec{S} \\ &= |\vec{F}| |\vec{S}| \cos \theta\end{aligned}$$

$$\text{Work Done} = |\vec{F}| |\vec{S}| \cos \theta$$



condition depends on

- 1) if $\theta = 0$, the displacement is along the direction of the force,

$$\therefore \underline{W = F \times S}$$

- 2) if $\theta = \pi$; that is the displacement is opposite direction of force,

Work is negative and work is ~~set~~ said to be done

against the force.

Let, u be the initial velocity
at 0 where acceleration

and v be the velocity at u .
where, acceleration

$$\therefore 550Hg = m \cdot \frac{dv}{du} \cdot u = m \cdot \frac{1}{n} \cdot v \cdot u$$

$$du = v/n$$

$$\Rightarrow v = nu - k$$

The eqn of motion is:

$$m \cdot \frac{dv}{dt} = \rho F = \frac{550Hg}{v} \cdot v$$

$$m \int_u^v \frac{dv}{v} = 550Hg \int_0^t dt$$

$$m \left[\ln v \right]_u^v = 550Hg t$$

$$\Rightarrow \frac{m}{2} [v^2 - u^2] = 550Hg \cdot t$$

$$\Rightarrow t = \frac{m}{2} \frac{v^2 - u^2}{550Hg}$$

$$\Rightarrow t = \frac{m(v^2 - u^2)}{1100Hg}$$

$$\Rightarrow t = \frac{m}{1100Hg} [n^2 u^2 + u^2]$$

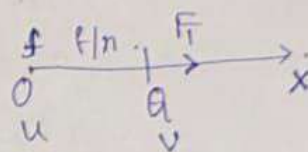
$$= \frac{m u^2 (n^2 + 1)}{1100Hg}$$

Q/ A mass of m pounds moves initially with velocity u foot/second along a straight line. A constant power $P = H \text{ Hg}$ is applied to it to increase its velocity. So that, the time that laps before the acceleration is reduced to $\frac{1}{n^2}$ of its original value H .

$$\frac{m}{1100 \text{ Hg}} (n^2 - 1) u^2$$

→ Let, O be the initial position of particle

Let, t time



Q be the position of particle of mass m at Q , $QO = x$, v is its velocity at u

Let, f be the initial acceleration after time t , its acceleration is f/n .

Now, apply $\Rightarrow P = F \cdot v$

$$\Rightarrow 550 \text{ Hg} = F \cdot v \quad \text{--- (1)}$$

For maximum velocity v

$$\frac{dv}{dt} = 0$$

$$F - Kv^2 = 0$$

$$\Rightarrow F = Kv^2$$

Therefore,

$$K = F/v^2$$

$$2240M \frac{dv}{dt} = \frac{550Hg}{v} - \frac{550Hg}{v^3} \cdot v^2$$

$$= \frac{550Hg}{v} \left(1 - \frac{v^2}{v^2}\right)$$

$$= \frac{550Hg}{v} \left(\frac{v^2 - v^2}{v^2}\right)$$

$$\frac{dv}{dt} = \frac{550Hg}{2240Mv^3} (v^2 - v^2)$$

$$\Rightarrow \int_0^v \frac{dv}{v^3 - v^2} = \int_0^v \frac{550Hg}{2240Mv^3} dt$$

$$\Rightarrow \int_0^v \frac{550Hg}{2240Mv^3} dt = \frac{1}{2v} \left[\log \left| \frac{v+v}{v-v} \right| \right]_0^v$$

$$\Rightarrow t = \frac{2240Mv^2}{550Hg} \log \left[\frac{v+v}{v-v} \right]$$

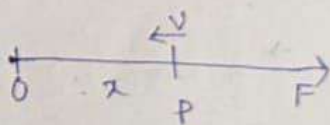
8) The horsepower required to a streamer of mass M Tons. At its maximum speed V feet/second. H. The resistance is proportional to square of the speed and the engine exerts a const. propeller thrust at all speed. If time in time t from rest, the streamer required velocity v feet/s. Then prove that,

$$t = \frac{112}{55} \frac{MV^2}{Hg} \log \left(\frac{V+v}{V-v} \right)$$

→ Let,

p be the position of streamer at time t .

•



Let, v is velocity at P .

Let, F be the poundal ~~be~~ the force.

Now,

$$P = Fv$$

$$550 Hg = F \cdot v$$

$$F = \frac{550 Hg}{v} \quad \text{--- (1)}$$

therefore, eqn of motion is

$$2240 M \frac{dv}{dt} = F - Kv^2$$

$$= \frac{1}{2} m \cdot 2gx$$

$$= mgx$$

~~P.E., P.E.~~

P.E. at P is

$$T = mg(h-x)$$

$$\begin{aligned} \therefore K + T &= mgx + mg(h-x) \\ &= mg(x+h-x) \\ &= mgh \end{aligned}$$

at the starting point, K.E = 0,
and P.E. (T) = mgh.

Therefore, $K + T = mgh = \text{const.}$

Again, the before impact from
the ground, $K = mgh$, $T = 0$

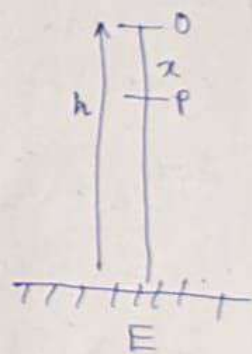
$$\therefore K + T = \underline{mgh}$$

* Principle of Energy:

The change in K.E of a body is equal to work done by the force.

- Q) For a particle of mass m falling from rest under gravity from a height h above the ground, the sum of the K.E and P.E of the particle is constant at every point of its path.

→ Let a particle fall freely under gravity from a point O at a height h above the earth surface E .



Let, v be the velocity of the particle, when it has fallen through a distance x .

$$\text{Then, } v^2 = 2gx$$

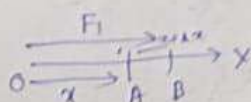
The K.E K at point P is

$$K = \frac{1}{2}mv^2$$

Date: 3.11.2023 MATH

Prove that, $P \Rightarrow \text{Power} = \text{Force} \times \text{velocity}$

$$P = FV$$



Let, a body move along the line Ox.

Let, A, B be the position of the particle at time $t, t + \Delta t$. Where $OA = x, OB = x + \Delta x$

Let, F be the force along the line AB. Therefore, $\Delta W =$ Elementary amount of the work doing in time Δt .

$$\Delta W = F \cdot \Delta x$$

P = rate of doing work

$$= \lim_{\Delta t \rightarrow 0} \frac{\Delta W}{\Delta t}$$

$$= \lim_{\Delta t \rightarrow 0} \frac{F \cdot \Delta x}{\Delta t}$$

$$= F \cdot \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t}$$

$$= F \cdot \frac{dx}{dt} = \boxed{F \cdot V}$$

$$y = a \cos \omega t + b \sin \omega t$$

$$\frac{d}{dt} (a \cos \omega t + b \sin \omega t)$$

$$= -a\omega \sin \omega t + b\omega \cos \omega t$$

$$\frac{d}{dt} = -a\omega^2 \cos \omega t + b\omega^2 \sin \omega t$$

$$= \omega^2 (-a \cos \omega t + b \sin \omega t)$$

$$= -\omega^2 (a \cos \omega t + b \sin \omega t)$$

$$= -\omega^2 y$$

$$\frac{d^2 y}{dt^2} + \omega^2 y = 0 \quad \left[\begin{array}{l} \text{this is the} \\ \text{simple} \\ \text{harmonic} \\ \text{motion} \end{array} \right]$$

Date: 1.12.2025

Force & Oscillation

Theorem

A particle moves in a straight line under the action of an acceleration directed towards a fixed pt in the line which varies as the distance

Q) A particle exist simple harmonic motion of period 10s and amplitude 5cm. Calculate maximum amplitude of velocity.

→ D $a = 5\text{cm}$, $T = 10\text{s}$
 $\hookrightarrow$ D

Angular frequency (ω) = $\frac{2\pi}{T}$

= $\frac{2 \times \frac{22}{7}}{10} \text{ s}^{-1}$

= $\frac{2\pi}{10} \text{ s}^{-1}$

Maximum velocity

= $a\omega$

= $0.5 \times \frac{2\pi}{10} \text{ s}^{-1}$

= $\pi \text{ cm/s}$

= 3.14 cm/s

Q) The displacement of a moving particle at any time t is given by $y = a \cos \omega t + b \sin \omega t$. So that, motion is simple harmonic.

$\frac{d^2x}{dt^2} = 0$

Simple harmonic motion

In other word, for the displacement
 $y = \pm a/\sqrt{2}$ from the mean of
position $P.E = \frac{1}{2} m$

from (1) we get

$$\begin{aligned} P.E &= \frac{1}{2} m \omega^2 a^2 \cos^2 \omega t \\ &= \frac{1}{2} m \omega^2 \frac{a^2}{2} \\ &= \frac{1}{4} m \omega^2 a^2 \end{aligned}$$

$$\begin{aligned} K.E &= \frac{1}{2} m \omega^2 (a^2 - y^2) \quad [\text{from (1)}] \\ &= \frac{1}{2} m \omega^2 \left(a^2 - \frac{a^2}{2} \right) \\ &= \frac{1}{2} m \omega^2 \left(\frac{2a^2 - a^2}{2} \right) \\ &= \frac{1}{4} m \omega^2 a^2 \end{aligned}$$

$$\text{Energy} = \frac{1}{2} m \omega^2 a^2$$

the K.E of a sh.
is given by

$$K.E = \frac{1}{2} m v^2$$

$$= \frac{1}{2} m \left(\frac{dy}{dt} \right)^2$$

$$= \frac{1}{2} m (-a\omega \sin \omega t)^2$$

$$= \frac{1}{2} m a^2 \omega^2 \sin^2 \omega t$$

$$= \frac{1}{2} m a^2 \omega^2 (1 - \cos^2 \omega t)$$

$$= \frac{1}{2} m \omega^2 (a^2 - \cos^2 a \omega t)$$

$$= \frac{1}{2} m \omega^2 (a^2 - y^2) \quad [\text{from (i)}]$$

$$P.E = \frac{1}{2} s y^2, \text{ where } s = \text{Stiffness}$$

$$\text{and } s/m = \omega^2$$

$$s = m \omega^2$$

$$P.E = \frac{1}{2} m \omega^2 y^2 \quad \text{--- (ii)}$$

when,

$$K.E = P.E$$

$$\Rightarrow \frac{1}{2} m \omega^2 (a^2 - y^2) = \frac{1}{2} m \omega^2 y^2$$

$$\Rightarrow a^2 - y^2 = y^2$$

$$\Rightarrow 2y^2 = a^2$$

$$\Rightarrow y^2 = \frac{a^2}{2} \Rightarrow y = \pm \frac{a}{\sqrt{2}}$$

Q) A simple harmonic motion oscillator is characterised by $y = a \cos \omega t$.

Calculate the displacement at which ~~the~~ $K.E = P.E$

OR

At ~~at~~ what displacement from the ~~main~~ ^{mean} position the total energy of a simple harmonic oscillator is $\frac{1}{2} K.E$ and $\frac{1}{2} P.E$

→ Let, y be the displacement at which $K.E$ of simple harmonic oscillator = its potential energy
 $K.E = P.E$

now, $y = a \cos \omega t$ -- (1)

$$\frac{d}{dt} (a \cos \omega t)$$

$$= -\sin \omega t \cdot a \omega \sin \omega t$$

Q) A particle of mass 100g is placed in the field of potential $U = (5x^2 + 10) \text{ ergs/gm}$. Find the Frequency

given,
 $\rightarrow U = (5x^2 + 10) \text{ ergs/gm}$

$$m = 100 \text{ g}$$

$$F = - \frac{dU}{dx}$$

$$= - \frac{d}{dx} (5x^2 + 10)$$

$$= -10x$$

taking x as displacement,

$$F = m \cdot \frac{d^2x}{dt^2}$$

$$\Rightarrow -10x = m \cdot \frac{d^2x}{dt^2}$$

$$\frac{d^2x}{dt^2} = -10x/m$$

from (4) we get

$$m_1 g = \frac{\lambda b}{a}$$

$$m_1 g =$$

$$(x-b) = A \cos \sqrt{g/b} \cdot t + t_0$$

$$(x-b) = c \cos \sqrt{g/b} \cdot t + 0$$

which is the extension of the string at any instance, t ,

from the instance t when m_2 falls off.

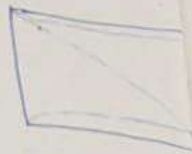
Hence, the depth of m_1

from the upper extremity of the string is $(a+x)$

$$(a+b+c \cos \sqrt{g/b} \cdot t) \quad [\text{proved}]$$

$$- A \left(\sin \sqrt{g/b} t + \epsilon \right)$$

$$x = -A \sin \left(\sqrt{g/b} t + \epsilon \right) - (vi)$$



Initially,

$$x = (b+c) \text{ and}$$

$$\dot{x} = 0 \text{ when}$$

$$t = 0$$

from (vi)

$$- A \sin \left(\sqrt{g/b} t + \epsilon_0 \right)$$

=

$$(x-b) = A \cos \left(\sqrt{g/b} t + \epsilon_0 \right)$$

$$b+c-b = 0 + \epsilon_0 A \cos \epsilon_0$$

$$c - \epsilon_0 A = A \cos \epsilon_0 \quad \text{--- (vii)}$$

and

from (vi)

$$- A \sin \left(\sqrt{g/b} t \right)$$

$$\Rightarrow - A \sqrt{g/b} \sin \epsilon_0 = 0$$

$$\epsilon_0 = 0 \text{ and } A = c \text{ [from (vii)]}$$

which shows that when both the masses are hanging together in equilibrium then the extension of the string is $(b+c)$

Now, let, m_2 falls off, then the mass m_1 will begin to rise.

Let, p be the position of m_1 at any time instance t

such that,

$$AP = x$$

$\therefore$ equⁿ of motion is

$$m_1 \ddot{x} = m_1 g - \lambda \frac{x}{a}$$

$$m_1 g - \frac{\lambda x}{a} = m_1 g - \frac{m_1 g x}{b}$$

$$= m_1 g \left[\frac{\lambda x}{a} = - \frac{m_1 g x}{b} \right]$$

[from (i)
 λ/a

$$\Rightarrow \ddot{x} = -\frac{g}{b} (x-b) \quad \text{--- (iv)}$$

the general solⁿ of (i) is

$$(x-b) = A \cos \sqrt{g/b} t + C_0 \quad \text{--- (v)}$$

and

$$\dot{x} = A \sqrt{g/b} \sin$$

Q) Two bodies M_1 and M_2 are attached to the lower end of an elastic string whose upper end is fixed and are hanging at rest; M_2 falls off.

Show that, the distance from M_1 from the upper end of the string at time t is $a + b + c \cos \sqrt{\frac{g}{b}} t$, where, a is natural length of string, b and c are the distance by which it would be extended when supporting M_1 and M_2 .

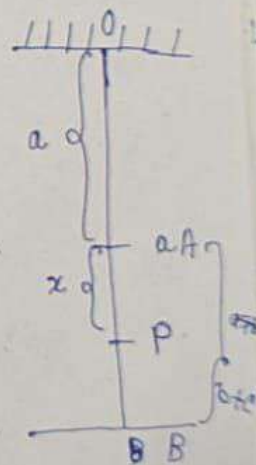
→ λ be the modulus of the elasticity of string

~~There~~
 $\therefore M_1 g = \lambda \frac{b}{a} \quad \text{--- (i)}$

$$m_2 g = \lambda \frac{b+c}{a} \quad \text{--- (ii)}$$

(i) + (ii)

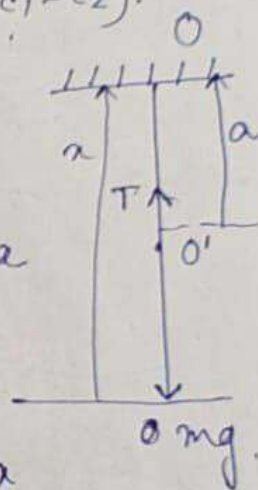
$$(m_1 + m_2) g = \frac{\lambda}{a} (b+c) \quad \text{--- (iii)}$$



Q) Let t_1, t_2 be the period of vertical oscillation of two different weight suspended by a elastic string and c_1, c_2 be the static extense due to ~~the~~ ^{these} weights prove that

$$g(t_1^2 - t_2^2) = 4\pi^2(c_1 - c_2).$$

→ Let, m_1, m_2 be the suspended from a point O by an elastic string of natural length a



Let, T_1, T_2 be the extension of string.

$$m_1 = \frac{\lambda c_1}{ag} \quad T_1 = \frac{\lambda c_1}{a} \quad \text{or} \quad m_1 g = \dots (1)$$

$$T_2 = \frac{\lambda c_2}{a} = m_2 g \dots (2)$$

at any instant,

$t =$

Let x be extension of the string.

After the blow, the new velocity is $\frac{3na}{2}$ and eqn of motion is,

$$v \frac{dv}{dx} = -n^2 x$$

$$\Rightarrow v^2 = -n^2 x^2 + C \quad \text{--- (3)}$$

from (3)

$$v = \frac{3na}{2}, \quad x = \frac{\sqrt{3}a}{2}$$

$$\Rightarrow \frac{9n^2 a^2}{4} = -n^2 \cdot \frac{3a^2}{4} + C$$

$$\Rightarrow \frac{9n^2 a^2}{4} + \frac{3n^2 a^2}{4} = C$$

$$\Rightarrow \frac{12n^2 a^2}{4} = C \quad C = \frac{3n^2 a^2}{1}$$

from (3)

$$\Rightarrow v = 0, \quad C = \frac{13n^2 a^2}{4}$$

$$0 = -n^2 x^2 + \frac{12n^2 a^2}{4} \quad 3n^2 a^2$$

$$n^2 x^2 = \frac{12n^2 a^2}{4}$$

$$x^2 = \frac{12n^2 a^2}{4n^2}$$

$$x^2 = \frac{12a^2}{4}$$

$$x = \frac{\sqrt{12} a}{2}$$

$$0 x^2 = \frac{3n^2 a^2}{n^2}$$

$$x = \sqrt{3} a$$

Since, $x=a, v=0$
from (1)

$$v^2 = -n^2 x^2 + A$$
$$= A = n^2 a^2$$

from (1) we get

$$v^2 = -n^2 x^2 + n^2 a^2$$
$$= n^2 (a^2 - x^2) \dots (2)$$

let, $v=u$, when $x = \frac{\sqrt{3}a}{2}$

from (2)

$$u^2 = n^2 \left(a^2 - \frac{3a^2}{4} \right)$$
$$= n^2 \left(\frac{4a^2 - 3a^2}{4} \right)$$

$$u^2 = n^2 \frac{a^2}{4}$$

$$u = \pm \frac{na}{2}$$

The new velocity is

$$\frac{na}{2} + na$$
$$= \frac{3na}{2}$$

Date: 19.11.2023

Q) A particle is oscillating in a straight line about a centre of force O. When a particle at a distance x from O, its acceleration is n^2x towards O and Amplitude is a .
 when a distance $\frac{\sqrt{3}a}{2}$ ~~is~~ from O, the particle receives a blow in the direction of motion which generates velocity na . If the velocity be away from O, so that the new amplitude is $\frac{\sqrt{3}a}{2}$.

→ given,

equⁿ of motion, $\ddot{x} = -n^2x$

$$\begin{aligned} \frac{d}{dt} v \cdot \frac{dv}{dx} &= -n^2x \\ &= \int v \cdot dv = \int -n^2x dx \end{aligned}$$

$$\Rightarrow \frac{v^2}{2} = -\frac{n^2x^2}{2}$$

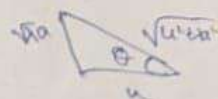
$$\Rightarrow v^2 = -n^2x^2 + A$$

--- (1)

to go from $a \rightarrow 0$ to 0 . when
 x varies from $A \rightarrow 0$, $a \rightarrow 0$

$$t_1 = - \int_0^a \frac{dx}{a \sqrt{u^2 + \lambda(a^2 - x^2)}}$$

$$= \int_0^a \frac{dx}{\sqrt{x^2 + \lambda a^2 - \lambda x^2}}$$



$$= \left[\frac{1}{\sqrt{\lambda}} \sin^{-1} \frac{\sqrt{\lambda} x}{\sqrt{x^2 + \lambda a^2 - \lambda x^2}} \right]_0^a$$

$$= \frac{1}{\sqrt{\lambda}} \sin^{-1} \frac{\sqrt{\lambda} a}{\sqrt{u^2 + \lambda a^2}}$$

$$= \frac{1}{\sqrt{\lambda}} \tan^{-1} \left(\frac{\sqrt{\lambda} a}{u} \right)$$

since, T be the period of time

$$\therefore T = \frac{2\pi}{\sqrt{\lambda}}$$

$$\Rightarrow \sqrt{\lambda} = \frac{2\pi}{T}$$

$$t_1 = T/2\pi \tan^{-1} \frac{2\pi a}{T \cdot u}$$

where, $OP = x$

The eqn of motion

$$\frac{d^2x}{dt^2} = -\lambda x - (1)$$

multiply both side by

$$2 \frac{dx}{dt} \text{ and}$$

integrating we get

$$\left(\frac{dx}{dt}\right)^2 = -\lambda x^2 + C_1 -$$

Integrating at

initially at $x = a$ and $\frac{dx}{dt} = u$

$$= u^2 = -\lambda a^2 + C_1$$

$$\rightarrow C_1 = u^2 + \lambda a^2$$

$$\left(\frac{dx}{dt}\right)^2 = -\lambda x^2 + C_1$$

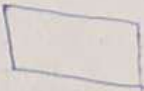
$$= -\lambda x^2 + u^2 + \lambda a^2$$

$$\frac{dx}{dt} = \sqrt{\lambda(a^2 - x^2) + u^2}$$

$$\Rightarrow \int_{-a}^0 \frac{dx}{\sqrt{u^2 + \lambda(a^2 - x^2)}} = \int_0^{t_1} dt$$

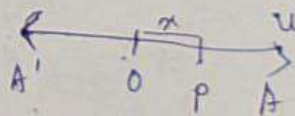
let, t_1 be the time taken by the particle

force $F \cos pt$ from book per unit mass

Q) A particle O is moving a ~~SHM~~ ^{SHM} of amplitude a and period T and when in a position of instantaneous rest is given a blow which imparts a velocity u  towards the mean centre. Show that it will be arrived at its next position of instantaneous rest at time $\frac{T}{\pi} \tan^{-1} \frac{uT}{2\pi a}$ than if it has not receive the impulse.

show that it will continue the SHM of the same period but of amplitude $\left(a^2 + \frac{v^2 T^2}{4\pi^2}\right)^{1/2}$

Let, O be the centre



of force, $OA = OA' = a$

be the amplitude of the motion.

Let, P be any pt. of line of motion at any time t .

$$= F \cdot e^{int} \cdot \frac{1}{(D+in)^2 + n^2} \cdot 1$$

$$= F \cdot e^{int} \cdot \frac{1}{D^2 + 2inD} \cdot 1$$

$$= F \cdot e^{int} \cdot \frac{1}{2inD} \left(1 + \frac{D}{2in}\right)^{-1}$$

$$= F \cdot e^{int} \cdot \frac{1}{2inD} \left(1 - \frac{D}{2in}\right) \cdot 1$$

$$= F \cdot e^{int} \cdot \frac{1}{2inD} \cdot 1$$

$$= \frac{F \cdot e^{int}}{2inD} \cdot 1$$

∴ its real part is

$$\frac{F \sin nt}{2n}$$

G.S is $x = a \cos(nt + \epsilon) + \frac{F}{2n} \sin nt$.

Damped Oscillation:

A particle is moving a st. line with an acceleration $n^2 \times (\text{distance})$ towards a fixed pt in the line, in a medium which offers a resistance $\propto$ velocity and is simultaneously acted on by a ~~dis~~ periodic disturbing

The motion of a particle is the
result can two
simple harmonic motion $[p \neq n]$
of a period $2\pi/n$, $2\pi/p$.

The 1st part is the usual
simple H.M. In the absence
of periodic disturbing force and
known as free oscillation.

And 2nd part is simply a Super
posed S.H.M and is known as
forced oscillation.

When $n^2 - p^2$ is very small, then the force
of oscillator. $\frac{F}{n^2 - p^2} \cos pt$. will be
very large then the particle will be
oscillate with a period which is
practically same as that of force
oscillation.

If, $[p = n]$ the P.I is the real part

$$\text{as } \frac{F}{D^2 + n^2} e^{int}$$

$$\frac{F}{D^2 + n^2} e^{int}$$

the complementary funⁿ is

$$C.F = A \cos nt + B \sin nt$$

$$= a \cos (nt + \epsilon) \quad \text{--- (11)}$$

(derivative 2 times for P.I)

The particular

Integral is.

$$A = a \cos \epsilon$$

$$B = a \sin \epsilon$$

$$P.I = \frac{1}{D^2 + n^2} F \cos pt$$

$$a (\cos \epsilon \cos nt - \sin \epsilon \sin nt)$$

$$= \frac{1}{\cancel{D^2 + n^2}} \left[D^2 = \frac{d^2}{dt^2} \right] \frac{a \cos (nt + \epsilon)}{}$$

$$= \frac{1}{-p^2 + n^2} F \cos pt$$

$$[p \neq n]$$

$$= \frac{F}{n^2 - p^2} \cos pt \quad \text{--- (111)}$$

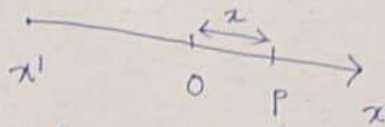
The general Solⁿ

$$\text{Q.1 } G.S = a \cos (nt + \epsilon) + \frac{F}{n^2 - p^2} \cos pt$$

$$[a, \epsilon \text{ are const.}]$$

from the fixed pt and simultaneously acted on by a periodic disturbing force $F \cos pt$ per unit mass.

→ Let, P be the position of the particle



At any time t moving along the line ox under the force mn^2x towards the fixed pt O . such that $OP = x$.

At the same time, it is also acted upon by a periodic force $mF \cos pt$.

Then, the equⁿ of motion is.

$$G.S = C.F + P.I \quad m\ddot{x} + mn^2x = mF \cos pt$$

$$\ddot{x} + n^2x = F \cos pt \quad \dots (1)$$

The general solⁿ of the equⁿ consist of a complimentary function and particular integral.

$$y = a \cos \omega t + b \sin \omega t$$

$$\frac{d}{dt} (a \cos \omega t + b \sin \omega t)$$

$$= -a\omega \sin \omega t + b\omega \cos \omega t$$

$$\frac{d}{dt} = -a\omega^2 \cos \omega t + b\omega^2 \sin \omega t$$

$$= \omega^2 (-a \cos \omega t + b \sin \omega t)$$

$$= -\omega^2 (a \cos \omega t + b \sin \omega t)$$

$$= -\omega^2 y$$

$$\frac{d^2 y}{dt^2} + \omega^2 y = 0 \quad \left[\text{this is the simple harmonic motion} \right]$$

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Force & Oscillation

Theorem

A particle moves in a straight line under the action of an acceleration directed towards a fixed pt in the line which varies as the distance